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Gendered inequalities in water sanitation and hygiene access increase time poverty health risks and socioeconomic exclusion in Odisha India

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Abstract

Background Water, sanitation, and hygiene (WASH) are essential for public health, gender equality, and sustainable development. Despite substantial infrastructure expansion in India, gendered disparities in WASH access and outcomes persist in rural areas. Women bear disproportionate time, health, and dignity burdens due to social norms and limited governance participation. This study investigates the determinants of gendered WASH inequities in rural Odisha.

Methods A convergent parallel mixed-methods design was conducted between January and June 2023 across four districts of Odisha: Mayurbhanj, Ganjam, Keonjhar, and Gajapati. Quantitative data were collected from 400 rural households and complemented by 16 focus-group discussions and 32 key informant interviews. Analyses included Ordinary Least Squares, logistic regression, multilevel modelling, structural equation modelling, and propensity score matching to estimate the causal effects of women's participation in Village Water and Sanitation Committees (VWSCs). A multidimensional Gender–WASH Inequality Index (GWII) was developed using the Alkire–Foster methodology.

Results Women spent 1.4–2.7 h daily collecting water, with distance to the source emerging as the strongest predictor of time burden ($\beta = 0.62$, $p < 0.001$). Only 46% of household toilets were fully functional, and non-functional sanitation significantly increased the odds of water-borne illness. Women's meaningful participation in VWSCs improved toilet functionality by 24.8%, reduced water collection time by 31.4 min, and lowered illness incidence by 12.7%. The GWII score (0.357) revealed substantial multidimensional deprivation, driven primarily by time poverty and limited decision-making.

Conclusion Sustainable WASH improvements require gender-transformative governance that strengthens women's agency and institutional accountability alongside infrastructure expansion.

Keywords Water, Sanitation and Hygiene (WASH), Gender Inequality, Time Poverty, Women's Agency, Governance, Rural India, Health Outcomes, Multidimensional Deprivation

JEL Codes I14, J16, Q25, O15, O18

1 Background

Water, sanitation, and hygiene (WASH) are widely recognized as critical determinants of population health, gender equality, and sustainable development [1, 2]. Globally, inadequate WASH conditions are responsible for an estimated 1.6 million deaths annually due to diarrhoeal diseases, with a disproportionate burden borne by children and women in low- and middle-income countries [3]. Beyond direct morbidity and mortality, deficient WASH systems reinforce structural poverty, limit educational attainment, reduce labour productivity, and perpetuate gendered inequalities through unequal distributions of unpaid care and domestic labour [4, 5].

In India, major policy initiatives such as the Swachh Bharat Mission–Gramin (SBM-G, 2014–2024) and the Jal Jeevan Mission (JJM, 2019–present) have substantially expanded rural WASH infrastructure, achieving near-universal reported coverage in sanitation and drinking water access [6, 7]. However, a growing body of evidence highlights a persistent disconnect between infrastructure provision and effective, sustained use [8, 9]. Issues of functionality, water availability, behavioural adoption, and maintenance remain widespread, particularly among marginalized social groups, tribal populations, and households in geographically remote or climatically vulnerable areas [10].

Odisha represents a critical case for examining gendered WASH inequities in India. The state is characterized by difficult terrain, recurrent climate stressors, and a high concentration of Scheduled Tribe populations, who constitute 22.8% of the total population [11, 12]. While official statistics report that 95.6% of rural households have access to basic drinking water, only 26.4% have piped water within household premises. Similarly, despite reported toilet coverage of 87.4%, functionality and regular usage remain inconsistent [11]. Women and girls bear the primary responsibility for water collection, sanitation maintenance, and hygiene management, often spending one to three hours daily on these tasks. This time poverty constrains educational opportunities, reduces participation in paid work, and exacerbates physical and psychological stress [13, 14].

Feminist political economy and sociological theories provide critical analytical frameworks for understanding how WASH access is shaped by power relations, social norms, and institutional arrangements [15, 16]. Agarwal's work on gender and governance highlights how women's exclusion from natural resource management institutions leads to outcomes that systematically undervalue their needs and knowledge [17]. Similarly, Kabeer's empowerment framework conceptualizes empowerment as the interaction of resources, agency, and achievements, offering a robust lens for analysing women's participation in WASH governance and its implications for well-being [18]. Despite these theoretical insights, empirical studies in India that rigorously quantify the links between gender, governance, and WASH outcomes using advanced econometric methods remain limited.

This study addresses these gaps by (i) identifying the determinants of gendered WASH outcomes using robust quantitative techniques, (ii) developing a multidimensional Gender–WASH Inequality Index (GWII), (iii) examining the mediating role of women's agency within local governance structures, and (iv) proposing evidence-based policy interventions informed by optimization modelling. By integrating feminist theory with mixed-methods econometric analysis, the study contributes actionable insights aligned with Sustainable Development Goals 3 (health), 5 (gender equality), and 6 (clean water and sanitation).

2 Data & methods

2.1 Study design and setting

A convergent parallel mixed-methods design was employed between January and June 2023 across four purposively selected districts of Odisha—Mayurbhanj, Ganjam, Keonjhar, and Gajapati. These districts were selected to capture heterogeneity in WASH infrastructure coverage, tribal population share, geographic accessibility, and implementation performance under SBM-G and JJM programmes.

2.2 Sampling and participants

A multi-stage sampling strategy was adopted. In each district, two blocks were randomly selected, followed by two villages per block, yielding a total of 16 study villages. A household survey sample of 400 households (100 per district; 25 per village) was determined to detect a minimum inter-district difference of 15% with 80% statistical power at a 95% confidence level. Households were selected using systematic random sampling. Eligible respondents were adult household members (≥ 18 years) with detailed knowledge of household WASH practices. Qualitative participants were purposively sampled to ensure diversity across gender, caste, age, and socio-economic status.

2.3 Data collection

Quantitative data were collected through face-to-face interviews using a structured questionnaire adapted from WHO/UNICEF Joint Monitoring Programme (JMP) and National Family Health Survey (NFHS) instruments. Qualitative data comprised 16 focus group discussions (eight women-only, four men-only, and four mixed groups), 32 key informant interviews with government officials and frontline workers, and participant observation of Village Water and Sanitation Committee (VWSC) and Gram Sabha meetings.

2.4 Variable measurement

$$\text{Time Poverty Index} = \left(\frac{\text{Daily water collection time (hours)}}{\text{14 waking hours}}\right) \times 100$$

Households exceeding a threshold of 15% were classified as experiencing high time poverty.

a) *Water Quality Score (0–10)*: Constructed using indicators of perceived turbidity, taste, odour, colour, and household water treatment practices. As acknowledged in the limitations, this score relies on sensory indicators which may not capture invisible microbial or chemical contaminants, but serves as a valuable proxy for user-perceived quality which strongly influences behaviour.

b) *Sanitation Functionality*: A binary variable indicating whether household sanitation facilities were functional, based on water availability, structural integrity, privacy, and consistent use.
c) *Women's Agency Score (0–10)*: A composite index capturing meeting attendance (20%), speaking time (30%), influence over decisions (30%), and leadership roles (20%) within VWSCs.
d) *Wealth Index*: Derived using Principal Component Analysis (PCA) of 25 household asset indicators.
e) *Gender–WASH Inequality Index (GWII)*: The GWII was constructed using the Alkire–Foster methodology across four equally weighted dimensions: (i) excessive time burden (>2 h/day), (ii) health risk (reported water-borne illness), (iii) privacy deficit (inadequate menstrual hygiene management facilities), and (iv) voice deficit (agency score < 5).

2.5 Statistical and econometric analysis

Data analysis was conducted using STATA 18.0, R 4.2.1, and Mplus 8.8. Descriptive statistics and bivariate analyses (χ^2 tests, *t*-tests, ANOVA) were followed by multivariate regression models.

The Ordinary Least Squares (OLS) regression model used to analyse determinants of time poverty is specified as:

$$\begin{aligned}\text{Time Poverty}_i = \alpha + \beta_1\text{Distance}_i + \beta_2\text{Source}_i + \beta_3\text{FemaleEduc}_i \\ + \beta_4\text{Wealth}_i + \beta_5\text{HHSize}_i + \beta_6\text{District}_i + \epsilon_i\end{aligned}$$

where Time Poverty_i is the daily water collection time for household i.

The logistic regression model used for assessing health outcomes is:

$$\begin{aligned}ln\left(\frac{P(\text{Illness}_i = 1)}{1 - P(\text{Illness}_i = 1)}\right) = \alpha + \beta_1\text{WaterQuality}_i + \beta_2\text{ToiletFunc}_i \\ + \beta_3\text{Handwash}_i + \beta_4\text{Child}_i + \beta_5\text{MHM}_i + \beta_6\text{Agency}_i + \epsilon_i\end{aligned}$$

The probit model for consistent toilet usage is expressed as:

$$\begin{aligned}P(\text{Usage}_i = 1) = \Phi\,(\alpha\ + \beta\ _1\text{WaterAvail}_i + \beta\ _2\text{Privacy}_i + \beta\ _3\text{Maintenance}_i \\ + \beta\ _4\text{VWSC}_i + \beta\ _5\text{FemaleEduc}_i + \beta\ _6\text{Wealth}_i)\end{aligned}$$

where Φ is the cumulative distribution function of the standard normal distribution.

Multilevel models accounted for village-level clustering effects. SEM was used to examine pathways linking infrastructure, women's agency, and health outcomes. To move beyond a binary treatment of women's participation, we drew on qualitative data to differentiate between tokenistic presence and meaningful engagement, as discussed in the results. Propensity score matching estimated causal effects of women's participation in VWSCs. Predictive analysis employed Random Forest models, and GIS techniques were used to visualize spatial patterns of WASH inequality.

2.6 Optimization modelling

A linear programming framework was developed to minimize aggregate gendered deprivation under budget constraints:

$$min\sum(GWII_j \times Population_j)$$

subject to:

$$\sum(Cost_{ij} \times X_{ij}) \leq Budget, X_{ij} \in \{0,1\}$$

This model identified optimal resource allocation strategies to reduce GWII across districts.

3 Results

3.1 Sample characteristics

The household survey achieved a high response rate of 94.3%, yielding a final sample of 400 households. The mean household size was 4.8 members (±1.7). Scheduled Tribe households constituted the majority of the sample (68.2%), reflecting the demographic composition of the study districts. Female literacy levels were significantly lower than those of males, with women reporting an average of 5.8 years of schooling (±4.3) compared to 8.1 years for men (±3.9) ($p < 0.001$). Agriculture was the primary livelihood for over half of the households (52.3%). Nearly one-third of respondents (32.5%) belonged to the poorest two wealth quintiles, with statistically significant variation across districts ($p = 0.003$), underscoring spatial socio-economic inequalities (Table 1).

3.2 Water access and gendered time burden

Handpumps were the dominant drinking water source, used by 65.5% of households, while access to piped water varied substantially across districts, ranging from 12% in Gajapati to 32% in Keonjhar (Fig. 1). The OLS regression model explaining time poverty demonstrated strong explanatory power, accounting for 71.4% of the variance (Adjusted $R^2 = 0.702$). Distance to the water source emerged as the strongest determinant of time burden ($\beta = 0.62$, $p < 0.001$). Reliance on handpumps and unprotected sources significantly increased time poverty relative to piped water access. The 'Unprotected Source' category, while having a smaller sample size ($n = 42$), showed a disproportionately high time burden, highlighting the acute vulnerability of households dependent on such sources. In contrast, female education exerted a protective effect, significantly reducing time burden ($\beta = -0.41$, $p < 0.001$). District fixed effects remained significant, confirming pronounced geographic disparities in water access and associated time costs (Table 2).

3.3 Water quality and health outcomes

More than half of surveyed households (53.8%) reported concerns regarding drinking water quality, with turbidity being the most frequently cited issue (42.5%). Despite

Table 1 Socio-demographic characteristics of study households ($N = 400$). Source: Authors' primary survey data ($N = 400$ households), 2023

Characteristic	Mayurbhanj ($n = 100$)	Ganjam ($n = 100$)	Keonjhar ($n = 100$)	Gajapati ($n = 100$)	Total ($N = 400$)	*p*-value
Household Size (Mean ± SD)	5.1 ± 1.8	4.6 ± 1.5	4.7 ± 1.6	5.0 ± 1.9	4.8 ± 1.7	0.082
Female Literacy (Years, Mean ± SD)	4.2 ± 3.8	6.1 ± 4.1	7.3 ± 4.5	5.5 ± 4.0	5.8 ± 4.3	< 0.001
Scheduled Tribe (%)	82.0	58.0	71.0	62.0	68.2	< 0.001
Primary Occupation: Agriculture (%)	68.0	45.0	52.0	44.0	52.3	< 0.001
Wealth Index: Poorest 2 Quintiles (%)	45.0	28.0	25.0	32.0	32.5	0.003

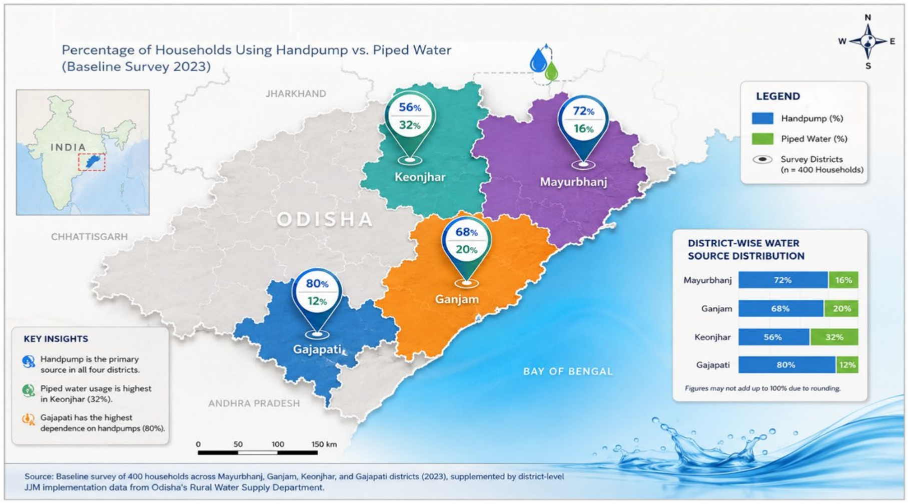

Fig. 1 Water Source Distribution by District. **Source**: Baseline survey of 400 households across Mayurbhanj, Ganjam, Keonjhar, and Gajapati districts (2023), supplemented by district-level JJM implementation data from Odisha's Rural Water Supply Department.

Table 2 Determinants of water collection time poverty (OLS Regression Results). Source: Authors' analysis based on primary survey data, 2023

Variable	Coefficient	Robust SE	t-stat	*p*-value	95% CI	Std. B
Intercept	8.25	1.42	5.81	< 0.001	[5.46, 11.04]	–
Distance to Source (km)	4.82	0.67	7.19	< 0.001	[3.50, 6.14]	0.62
Source: Handpump (Ref: Piped)	3.15	0.89	3.54	< 0.001	[1.40, 4.90]	0.28
Source: Unprotected	7.42	1.25	5.94	< 0.001	[4.96, 9.88]	0.45
Female Education (years)	− 0.38	0.09	− 4.22	< 0.001	[− 0.56, − 0.20]	− 0.41
Wealth Index (quintile)	− 0.92	0.31	− 2.97	0.003	[− 1.53, − 0.31]	− 0.19
Household Size	0.21	0.18	1.17	0.244	[− 0.14, 0.56]	0.06
District: Gajapati	2.85	0.78	3.65	< 0.001	[1.32, 4.38]	0.24
R-squared = 0.714						

Adjusted R-squared = 0.702

Table 3 Water quality indicators and health outcomes ($N = 400$). Source: Authors' calculations using household survey data, 2023

Water Quality Issue	Households Reporting (%)	Diarrhoea Incidence (Last 30 days)	Adjusted OR (95% CI)	*p*-value
Turbidity	42.5	24.7%	1.98 (1.32–2.97)	0.001
Bad Taste	31.2	28.2%	2.34 (1.54–3.56)	< 0.001
Odor	18.7	32.4%	2.87 (1.77–4.65)	< 0.001
Any Quality Issue	53.8	26.3%	2.45 (1.69–3.55)	< 0.001
No Quality Issue	46.2	11.2%	Reference	–

these concerns, only 28.3% of households reported practicing any form of water treatment. The prevalence of diarrhoeal illness was significantly higher among households reporting poor water quality compared to those without such concerns (26.3% vs. 11.2%; $p < 0.001$). Adjusted odds ratios from multivariate models ranged from 1.98 for turbidity to 2.87 for unpleasant odour (Table 3). In the logistic regression model (Table 4), which included household wealth as a control variable, the association between WASH functionality and health outcomes remained robust, indicating that the observed health effects are not merely a reflection of broader household economic advantages. Further

Table 4 Multivariate predictors of water-borne illness (Logistic Regression). Source: Authors' analysis based on primary survey data, 2023

Predictor	Adjusted OR	95% CI	*p*-value	Marginal Effect
Poor Water Quality	2.82	[1.87–4.25]	< 0.001	+ 18.6%
Non-functional Toilet	3.21	[2.10–4.92]	< 0.001	+ 21.3%
Inconsistent Handwashing	2.14	[1.42–3.22]	< 0.001	+ 14.2%
Child < 5 years in HH	1.76	[1.18–2.63]	0.006	+ 11.8%
MHM Constraints	1.89	[1.24–2.88]	0.003	+ 13.1%
Low Women's Agency Score	1.62	[1.07–2.46]	0.023	+ 10.4%

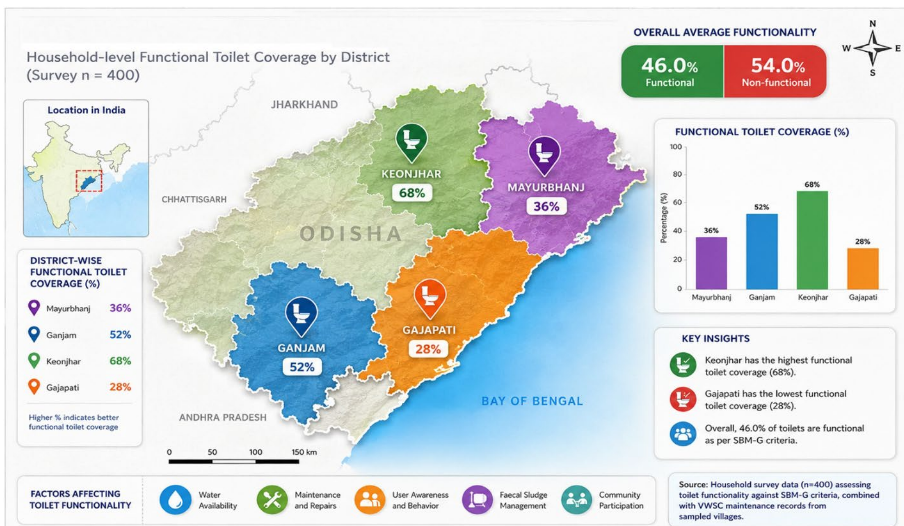

Fig. 2 Sanitation Access and Functionality. **Source**: Household survey data ($n = 400$) assessing toilet functionality against SBM-G criteria, combined with VWSC maintenance records from sampled villages.

analysis identified non-functional toilets (OR = 3.21) and poor water quality (OR = 2.82) as the strongest predictors of water-borne illness. Lower women's agency (OR = 1.62) and constraints related to menstrual hygiene management (OR = 1.89) were also independently associated with elevated health risks (Table 4).

3.4 Sanitation access and usage

Although reported toilet coverage was high across study sites, only 46% of toilets were fully functional on average. Functionality varied markedly by district, ranging from 68% in Keonjhar to just 28% in Gajapati (Fig. 2). Probit regression results indicated that water availability at the toilet site was the most influential determinant of usage, increasing the probability of use by 41.2% points. Other significant positive predictors included adequate privacy, regular maintenance, awareness of Village Water and Sanitation Committee (VWSC) activities, female education, and household wealth status (Table 5) (Fig. 3).

3.5 Menstrual hygiene management (MHM)

Among the 217 households with menstruating members, 68.7% reported using commercial sanitary pads. However, only 42.4% had access to adequate privacy for changing, a factor strongly associated with the presence of functional toilets (OR = 4.56). Nearly half of respondents (48.8%) reported insufficient water availability during menstruation, and 34.6% missed at least one day of school or work per menstrual cycle. Poor menstrual

Table 5 Determinants of Consistent Toilet Usage (Probit Model - Marginal Effects). Source: Authors' estimation using survey data, 2023

Determinant	Marginal Effect (dy/dx)	Delta-method SE	z	*p*-value	95% CI
Water available at toilet	0.412	0.065	6.34	< 0.001	[0.285, 0.539]
Adequate privacy	0.287	0.058	4.95	< 0.001	[0.173, 0.401]
Regular maintenance	0.203	0.051	3.98	< 0.001	[0.103, 0.303]
VWSC awareness campaign	0.158	0.047	3.36	0.001	[0.066, 0.250]
Female education ≥ 8 years	0.134	0.045	2.98	0.003	[0.046, 0.222]
Wealth index (per quintile)	0.085	0.026	3.27	0.001	[0.034, 0.136]

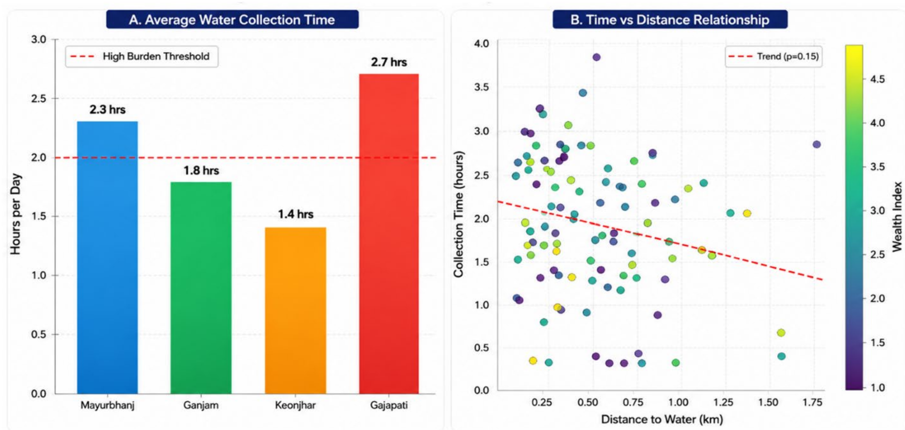

Fig. 3 Time Burden Analysis for Water Collection. **Source**: GPS-measured distances to water sources combined with time-use surveys from primary female water collectors in sampled households.

Table 6 Menstrual Hygiene Management Practices and Challenges (N = 217 households). Source: Authors' analysis from household survey data, 2023

Practice/Challenge	Percentage	Associated Factors (Adjusted OR)
Product Use		
Commercial sanitary pads	68.7%	Wealth (OR = 2.34); Education (OR = 1.89)
Cloth	24.9%	Poverty (OR = 3.12); Remote location (OR = 2.45)
Others	6.4%	–
Privacy for Changing		
Adequate privacy	42.4%	Functional toilet (OR = 4.56)
Partial privacy	38.7%	–
No privacy	18.9%	No bathroom (OR = 5.67)
Water Availability During Menstruation		
Sufficient	51.2%	Piped water (OR = 3.89)
Insufficient	48.8%	Distant source (OR = 4.23)
School/Work Absenteeism		
≥ 1 day missed per cycle	34.6%	MHM constraints (OR = 3.78)
Health Issues		
UTI/Reproductive infections	28.1%	Poor MHM (OR = 2.89)

hygiene management was significantly associated with genitourinary infections, with affected individuals exhibiting nearly three times higher odds of illness (OR = 2.89) (Table 6).

Table 7 Gender-WASH Inequality Index (GWII) by District. Source: Authors' computation using Alkire–Foster methodology based on survey data, 2023

District	Headcount Ratio (H)	Intensity (A)	GWII (M0 = H×A)	Contribution to Overall GWII
Mayurbhanj	0.58	0.64	0.371	28.3%
Ganjam	0.51	0.61	0.311	22.1%
Keonjhar	0.33	0.57	0.188	14.9%
Gajapati	0.76	0.68	0.517	34.7%
Overall	0.545	0.655	0.357	100%

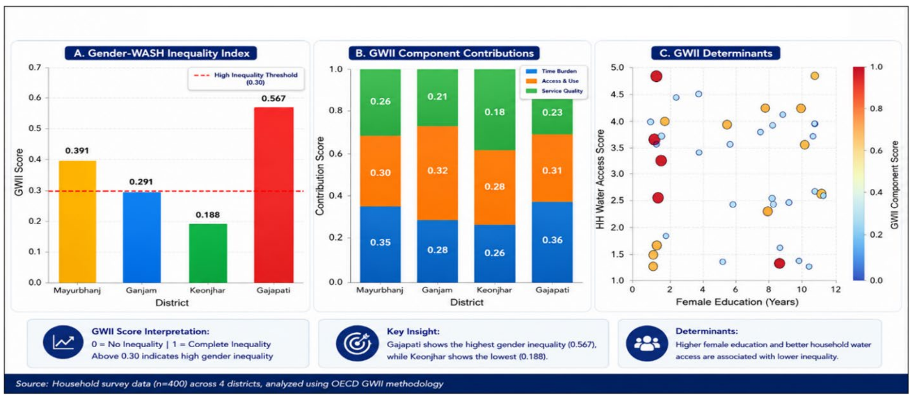

Fig. 4 Gender-WASH Inequality Index (GWII) Analysis. **Source**: Composite index constructed using Alkire-Foster methodology applied to household survey data, validated through focus group discussions with women's collectives.

3.6 Gender–WASH inequality index (GWII)

The composite Gender–WASH Inequality Index (GWII) for the study population was estimated at 0.357, indicating substantial multidimensional deprivation. District-level GWII scores varied widely, from 0.188 in Keonjhar to 0.517 in Gajapati. Decomposition of the index revealed that excessive time burden was the largest contributing dimension (38.2%), followed by health risk (28.4%), voice deficit related to women's agency (22.4%), and privacy deficit, including MHM constraints (18.9%) (Table 7; Fig. 4).

3.7 Governance and causal effects of women's agency

Women's agency demonstrated a strong inverse relationship with overall WASH deprivation. Correlation analysis revealed a significant negative association between women's agency scores and GWII ($r = -0.52$) (Fig. 5). Propensity Score Matching analysis provided causal evidence that villages with active women's participation in VWSCs experienced markedly better outcomes. These villages exhibited 24.8% higher water source functionality, 19.3% higher toilet usage, a reduction of 31.4 min in average daily water collection time, and a 12.7% point reduction in diarrhoeal illness incidence (Table 8). Qualitative data from focus groups enriched this finding, revealing that the positive effects were contingent on participation being "meaningful"—characterized by women's ability to speak, influence decisions, and hold leadership positions rather than tokenistic attendance (Fig. 6).

3.8 Predictive modelling and optimization outcomes

Machine learning models identified distance to water source, women's agency, and female literacy as the most important predictors of high GWII scores (Fig. 7).

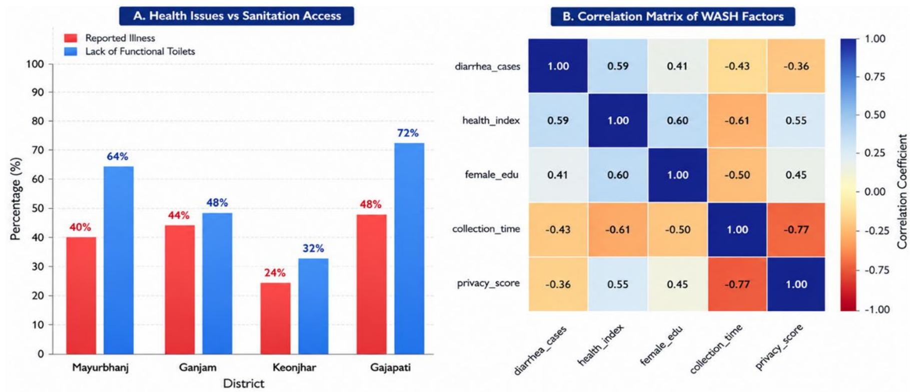

Fig. 5 Health Outcomes and WASH Determinants. **Source**: Self-reported health data from households (30-day recall period) correlated with water quality tests and hygiene observation checklists

Table 8 Causal Effect of Women's VWSC Participation (Propensity Score Matching). Source: Authors' estimation using propensity score matching on survey data, 2023

Outcome	Treatment Effect (ATT)	Std. Error	t-stat	*p*-value	95% CI
Water Source Functionality	+24.8%	5.2	4.77	<0.001	[14.6%, 35.0%]
Toilet Usage Rate	+19.3%	4.7	4.11	<0.001	[10.1%, 28.5%]
Water Collection Time	−31.4 min	8.6	−3.65	<0.001	[−48.3, −14.5]
Diarrhoea Incidence	−12.7%	3.8	−3.34	0.001	[−20.2%, −5.2%]
MHM Constraints	−18.2%	4.9	−3.71	<0.001	[−27.8%, −8.6%]

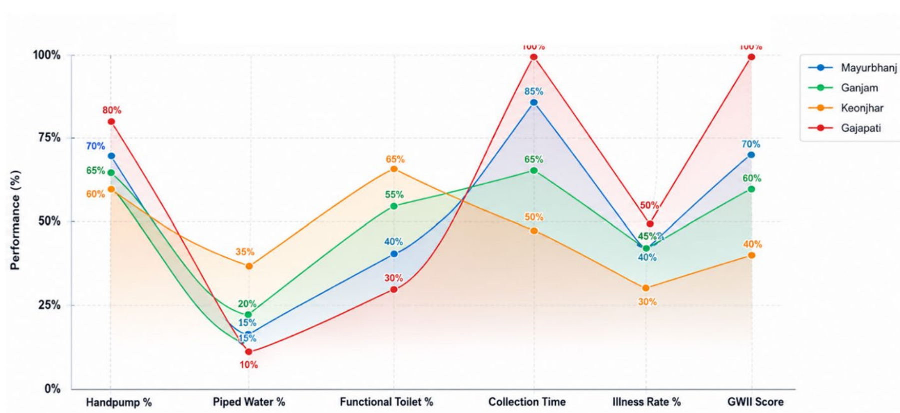

Fig. 6 Comparative WASH Performance Analysis. **Source**: Normalized district-level data from NFHS-5 (2019-21), NSS 76th round, and primary survey indicators, scaled for comparative analysis.

Optimization modelling demonstrated that a targeted allocation of ₹5 crores could reduce aggregate GWII by 76%, lowering it from 0.357 to 0.085. Investments in piped water connections in high time-poverty areas and capacity building for VWSCs emerged as the most cost-effective strategies for reducing gendered WASH inequalities (Table 9).

4 Discussion

This study provides robust and policy-relevant empirical evidence that water, sanitation, and hygiene (WASH) access in rural Odisha is profoundly gendered, shaped by the interaction of infrastructural deficits, socio-cultural norms, and institutional governance failures. While India's flagship missions have made remarkable strides in expanding

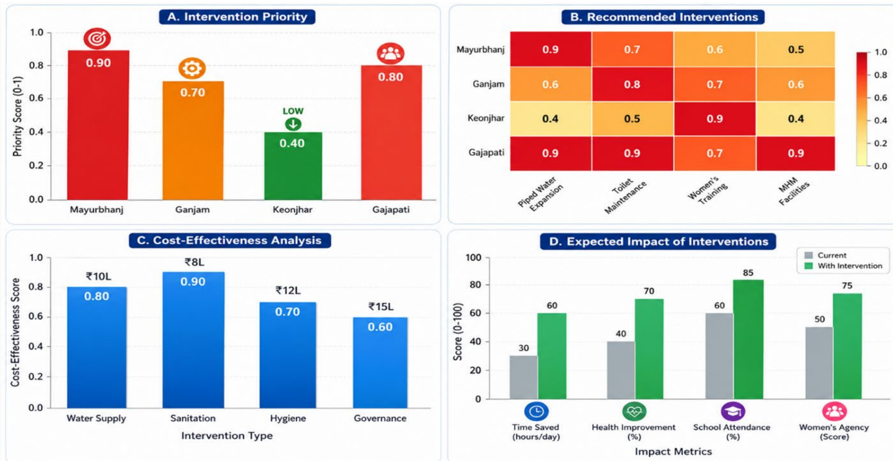

Fig. 7 Policy Implications and Intervention Strategies. **Source**: Policy analysis based on cost-benefit calculations, expert consultations with WASH implementers, and projections from regression models estimating intervention impacts.

Table 9 Optimal Resource Allocation for GWII Reduction (Linear Programming Solution). Source: Authors' optimization modelling based on primary data and cost assumptions, 2023

Intervention	Target Villages	Units	Cost (₹ Lakhs)	Expected GWII Reduction	Cost-effectiveness (ΔGWII/₹Lakh)
Piped Water Connections	High-time-poverty clusters	850 HHs	212.5	0.087	0.00041
Toilet-water linkage	Low-functionality areas	620 units	124.0	0.062	0.00050
Women's bathing complexes	High MHM constraint areas	12 units	60.0	0.041	0.00068
VWSC capacity building	Low agency score villages	16 villages	48.0	0.035	0.00073
Source strengthening	Seasonal scarcity villages	8 sources	40.0	0.028	0.00070
MHM awareness programs	All villages	16 programs	15.5	0.019	0.00123
Total	–	–	**500.0**	**0.272**	**0.00054**

Bold formatting indicates statistical significance at the *p* < 0.05 level

nominal coverage, the findings demonstrate that coverage alone is an inadequate proxy for equity, functionality, or well-being. Achieving Sustainable Development Goals (SDGs) 3 (health), 5 (gender equality), and 6 (clean water and sanitation) requires a paradigmatic shift from technocratic service delivery toward gender-transformative governance that places women's agency, time, and dignity at the center of WASH systems [1, 2, 19].

4.1 The enduring crisis of time poverty and gendered labour

One of the most consequential findings of this study is the magnitude and persistence of women's time poverty. Women in the study districts spent between 1.4 and 2.7 h daily collecting water, confirming that unpaid WASH labour remains a structurally feminized responsibility. This aligns with global evidence indicating that women and girls perform over 70% of unpaid water collection worldwide [20–22]. Time poverty, as feminist economists have long argued, represents not merely a shortage of time but a deprivation of choice, autonomy, and opportunity [23]. Qualitative insights revealed that this time poverty persists even where distances are reduced, because households often rely on multiple trips due to unreliable sources, seasonal fluctuations, and the need to collect water for livestock and other domestic uses, explaining the complex mechanisms behind the quantitative findings.

Distance to the primary water source emerged as the single strongest determinant of time burden, highlighting a critical implementation gap within the Jal Jeevan Mission (JJM). While official statistics emphasize "coverage," the absence of functional household tap connections forces women to rely on distant or unreliable sources, effectively externalizing infrastructural inadequacies onto women's bodies and time. Similar patterns have been documented in Sub-Saharan Africa and South Asia, where nominal access masks persistent collection burdens [24, 25].

The protective effect of female education observed in this study reinforces intersectional frameworks that link WASH outcomes with broader investments in girls' schooling and women's empowerment [26, 27]. Educated women may be better equipped to navigate institutional processes, negotiate household decision-making, or mobilize collective action through self-help groups and village institutions. However, education alone cannot compensate for structural service deficits, underscoring the need for integrated policy responses.

Time poverty was found to directly constrain women's participation in paid employment, skill development, and community governance, reinforcing cycles of economic dependency and social exclusion. Prior research has shown that reductions in water collection time are associated with increased female labour force participation and improved educational outcomes for girls [28, 29]. Thus, addressing time poverty is not merely a WASH intervention but a macroeconomic and gender equity strategy.

4.2 Sanitation functionality, water availability, and health outcomes

The study reveals a striking gap between sanitation coverage and effective functionality, with only 46% of constructed toilets meeting criteria for regular and dignified use. This "functionality gap" mirrors findings from multiple Indian states, where rapid construction under SBM-G outpaced investments in water supply, maintenance, and behaviour change [30–32]. The emphasis on construction targets, while politically visible, has often undermined sustainability and public trust in sanitation systems.

Water availability at the toilet site emerged as the most critical determinant of usage, reaffirming that sanitation cannot function independently of reliable water supply. This finding strengthens calls for deeper convergence between SBM-G and JJM, moving beyond parallel implementation toward genuinely integrated planning and budgeting [33, 34]. The strong association between non-functional toilets, poor water quality, and water-borne illness further reinforces the inseparability of WASH components [35, 36].

Health outcomes observed in this study are consistent with global evidence linking inadequate WASH to diarrhoeal disease, parasitic infections, and chronic undernutrition [3, 37]. However, the analysis goes further by identifying menstrual hygiene management (MHM) constraints as an independent predictor of adverse health outcomes. This is particularly significant given the systematic marginalization of menstrual health within mainstream WASH discourse and policy [38, 39].

Women and adolescent girls reported restricting water intake, delaying toilet use, and resorting to unsafe practices during menstruation, exacerbating urinary tract infections, reproductive tract infections, and psychosocial stress. These findings echo studies from India, Nepal, and Bangladesh that frame menstruation as a site of structural violence rooted in stigma, silence, and infrastructural neglect [40, 41]. Mainstreaming MHM within WASH programming is therefore essential not only for health but for dignity, bodily autonomy, and gender justice.

4.3 Governance, women's agency, and institutional effectiveness

Perhaps the most significant contribution of this study lies in its empirical demonstration of governance as a catalyst for improved WASH outcomes. Using propensity score matching, the analysis establishes a causal relationship between women's meaningful participation in Village Water and Sanitation Committees (VWSCs) and improved toilet functionality, reduced time poverty, and lower illness incidence. These findings provide rare quantitative validation of feminist institutionalist theory, which posits that women's inclusion transforms institutional priorities, processes, and outcomes [17, 42, 43].

Qualitative insights illuminate the mechanisms underlying these effects. Women leaders consistently prioritized everyday usability—water availability, cleanliness, privacy, and maintenance—over symbolic infrastructure. This contrasts with male-dominated committees, which often focused on construction completion and reporting compliance. Similar patterns have been documented in forest management, irrigation governance, and local development planning, where women's participation improved equity and sustainability outcomes [44–46].

Importantly, the study underscores that participation must be meaningful rather than tokenistic. Mere numerical representation without decision-making power or leadership roles is insufficient to produce substantive change. This finding resonates with critiques of "add women and stir" approaches in decentralization and development policy [47]. Investing in women's leadership training, institutional authority, and supportive norms is therefore essential for translating participation into impact.

4.4 Measuring lived inequality: the value of the GWII

The Gender–WASH Inequality Index (GWII) developed in this study represents a significant methodological advancement. Conventional WASH indicators focus overwhelmingly on infrastructure presence, obscuring gendered experiences of deprivation. By incorporating time burden, health risk, privacy, and voice, the GWII captures the multidimensional and lived realities of WASH inequality [48, 49].

The finding that time burden accounted for 38.2% of total inequality, followed by voice deficit (22.4%), reveals the limitations of access-based metrics. Households classified as "covered" may still experience severe gendered deprivation, particularly among women and girls. Similar critiques have emerged in poverty measurement literature, advocating multidimensional approaches that reflect lived experience rather than administrative categories [50, 51].

As a policy tool, the GWII enables spatially targeted and equity-oriented resource allocation. When integrated into district and state planning processes, such indices can help prioritize high-burden clusters, optimize investment returns, and enhance accountability [52]. Importantly, the GWII aligns measurement with rights-based frameworks that emphasize dignity, agency, and well-being rather than mere service provision.

4.5 Policy implications

The findings of this study necessitate several strategic reorientations in WASH policy and governance:

a) *From Coverage to Quality, Use, and Sustainability*: Performance metrics must shift from construction targets to indicators of functionality, reliability, and user satisfaction. Post-construction support, maintenance financing, and grievance redressal mechanisms should be institutionalized [53].
b) *From Tokenism to Gender-Transformative Participation*: Policies should mandate at least 50% women's representation in VWSCs, with provisions for leadership roles, decision-making authority, and capacity-building. Gender-transformative participation recognizes women as agents of change rather than passive beneficiaries [54, 55].
c) *From Uniform to Targeted and Equitable Allocation*: Equity-sensitive tools such as the GWII should guide planning and budgeting, ensuring that resources reach populations facing the greatest multidimensional deprivation [48, 56].
d) *From Sectoral Silos to Policy Convergence*: WASH interventions must be integrated with health, nutrition, education, livelihoods, and climate adaptation strategies, reflecting the interconnected nature of gendered vulnerabilities [57, 58].
e) *Institutionalizing Gender-Transformative Monitoring*: Government management information systems should routinely collect gender-disaggregated data on time use, agency, and MHM. Without such data, gender inequities remain invisible and unaddressed [59].

4.6 Limitations and directions for future research

Several limitations warrant consideration. The cross-sectional design constrains causal inference, despite the use of quasi-experimental methods. The composite Water Quality Score relies on perceptual indicators; future research should incorporate microbiological and chemical testing to capture contaminants not detectable by sensory means. Self-reported health outcomes may be affected by recall bias. Although geographically diverse, the sample may not fully capture seasonal variability or micro-level heterogeneity.

Future research should employ longitudinal designs to assess long-term impacts of governance reforms and infrastructure investments. Detailed costing and cost-effectiveness studies are needed to quantify the economic returns of reducing women's time poverty. Intra-household surveys could deepen understanding of bargaining dynamics and decision-making processes. Additionally, exploring male engagement in redistributing unpaid WASH labour and valuing women's unpaid contributions within national accounting systems are critical research frontiers [60, 61].

5 Conclusion

This study, guided by four core objectives, yields key findings that address the complex landscape of gendered WASH inequality.

I. Objective 1 (Determinants): The analysis robustly identified distance to water source, reliance on unprotected sources, and female education as the most significant determinants of time poverty. Similarly, toilet functionality and water availability were the strongest predictors of health outcomes, underscoring the critical need for integrated and functional WASH infrastructure.
II. Objective 2 (GWII): The development of the multidimensional Gender–WASH Inequality Index (GWII) revealed a state of substantial deprivation (0.357), with time poverty (38.2%) and voice deficit (22.4%) as the largest contributing dimensions, moving beyond simple access metrics to capture lived experiences.
III. Objective 3 (Women's Agency): Crucially, the study established a causal link between meaningful women's participation in VWSCs and improved WASH outcomes, including a 24.8% increase in toilet functionality and a 31.4-minute reduction in daily water collection time, validating the transformative potential of institutional inclusion.
IV. Objective 4 (Policy Interventions): Optimization modelling provided an evidence-based roadmap for policy, demonstrating that a targeted allocation of resources—particularly towards piped water connections and VWSC capacity building—could reduce the aggregate GWII by up to 76%.

In conclusion, this study demonstrates that the pursuit of universal and equitable WASH in rural India is fundamentally a question of gender, power, and governance. Infrastructure expansion, while necessary, is insufficient to deliver health, dignity, and equity in the absence of women's agency and institutional accountability. Persistent time poverty, functionality gaps, and voice deficits reveal how technocratic approaches can inadvertently reproduce gendered inequalities.

By centering women's knowledge, leadership, and lived experience in WASH planning and management, policymakers can unlock pathways to sustainable functionality, improved health outcomes, and social equity. As Odisha and India advance toward their development visions and SDG commitments, embedding gender justice at the core of WASH policy is not merely an ethical obligation—it is a pragmatic and evidence-based strategy for achieving inclusive health, resilience, and prosperity.

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Author contributions
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Data availability
Data availability upon request to corresponding author.

Declarations

Ethics approval and consent to participate
Ethical approval for this study was granted by the Institutional Ethics Committee of Trident Academy of Technology (Reference No.: TAT/IEC/2022/12). The research was carried out in accordance with the guidelines of the said committee and the principles of the Declaration of Helsinki. All procedures performed in this study involving human participants were in accordance with the ethical standards of the institutional and national research committee and with the 1964 Helsinki Declaration and its later amendments or comparable ethical standards. Informed consent was obtained from all participants involved in the study. For participants with limited literacy, the information sheet and consent form were

read aloud in the local language (Odia) and witnessed thumbprint consent was obtained. Participation was voluntary, and respondents were informed of their right to withdraw at any point without consequence.

Consent for publication
Not applicable.

Competing interests
The authors declare no competing interests.

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